

Immunomodulatory drug mechanisms

Part 1 | Immunomodulators act at different biological levels

A signalling inhibitor, an antimetabolite and a cell-targeting antibody are not interchangeable mechanisms.



T-cell signalling and B-cell surface targeting are shown as different intervention levels.

A causal readout

Reduced calcineurin-dependent signalling and IL-2 transcription identify a T-cell activation pathway. Reduced nucleotide synthesis points to a metabolic target. CD20-positive B-cell depletion identifies a surface-antigen population; it does not imply direct destruction of every existing antibody molecule.

A pathway assay can distinguish related drugs

The inhibited biochemical target matters more than the generic label steroid-sparing.

Mechanism

Feature	Target level	Expected primary readout
Calcineurin inhibition	T-cell signalling	Reduced IL-2 transcription
mTOR inhibition	Growth/proliferation signalling	Changed downstream growth response
Antimetabolite action	Nucleotide/metabolic pathway	Reduced substrate-dependent proliferation

A selective experiment

If guanine synthesis and mTOR activity remain demonstrably intact while calcineurin signalling falls, use those controls to distinguish the pathway. Similar clinical anti-inflammatory effects do not prove that the drugs interrupt the same intracellular step.

Target-antigen distribution predicts spared cells

A cell's developmental stage can determine whether a surface antibody can bind it.

Rituximab model

Population	CD20 relationship
Mature B cells	Target-bearing population
Normal mature plasma cells	Generally lack CD20
Haematopoietic stem cells	Do not express the target

Functional consequence

Long-lived plasma cells can continue antibody production despite marked depletion of circulating B cells. Existing antibodies also persist according to their turnover. Measure cell populations and circulating immunoglobulin separately; neither is a perfect substitute for the other.

A mechanism does not define every safety decision

Monitoring needs the drug's relevant tissue effects and the patient's circumstances.

Interpretation

Feature	What to assess	Why
Cell depletion	Infection history and relevant screening	Host defence may be altered
Metabolic inhibition	Blood count and organ function	Toxicity can affect renewing tissues
Signalling inhibition	Drug-specific renal/BP issues	Tissue effects differ by agent

Scope of these notes

These are mechanism categories, not interchangeable monitoring schedules. Use the individual product information and specialist protocol for a real regimen. A generic immune suppression label is insufficient to predict a particular contraindication, interaction or laboratory interval.

Rescue locates the block more precisely

An experimentally restored intermediate can distinguish upstream failure from downstream insensitivity.

Original pathway experiment

- Measure the inhibited signal
- ↓
- Confirm upstream activation is preserved
- ↓
- Restore the relevant intermediate or target
- ↓
- Reassess the downstream response

A limit on inference

A response to rescue supports a pathway contribution under those conditions; it does not establish unlimited clinical benefit. State the preserved and altered readouts explicitly. This is stronger reasoning than matching one symptom to a memorised drug name.